Author result prop-psi (printed 3.11) of ahl-163-stochastic-nonlinear-wave-dynamics-on-compact-2023

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- The statement and its own proof as printed in the article. source lines 1501-1605. the counted unit and the argument carried out in the span

Source reference. Original lemma 2.1: sources/ahl-163-stochastic-nonlinear-wave-dynamics-on-compact-2023/pinned-original-source.tex, source lines 547–547. Navigation pointer carrying the number the article prints (from census v3.14); nothing of the referenced item is copied.

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Proposition 0.1. *Let $0 < \varepsilon \ll 1$, $k \in \mathbb{N}$, $T \in (0;1]$ and $p, q \in [1,\infty)$. Then, $\{H_k(\mathbf{P}_N\Psi(\omega);\sigma_N(t,\mathbf{x}))\}_{N\in\mathbb{N}}$ is a Cauchy sequence in $L^p(\Omega;L^q([0,T];W^{-\varepsilon,\infty}(\mathcal{M})))$. In particular, denoting the limit by $:\Psi^k:$, we also have that $H_k(\mathbf{P}_N\Psi(\omega);\sigma_N(t,\mathbf{x}))$ converges almost surely towards $:\Psi^k:$ in $L^q([0,T];W^{-\varepsilon,\infty}(\mathcal{M}))$, and for $k=1$, we have that Ψ belongs almost surely to $C([0,T];W^{-\varepsilon,\infty}(\mathcal{M}))\cap C^1([0,T];W^{-\varepsilon-1,\infty}(\mathcal{M}))$. Moreover $H_k(\mathbf{P}_N\Psi(\omega);\sigma_N(t,\mathbf{x}))$, $:\Psi^k:$, and Ψ respectively obey the tail estimates* (3.6), (3.7), *and* (3.26).

Proof. As before, we can compute for fixed $t\in[0,T]$ and $\mathbf{x}\in\mathcal{M}$

$$\begin{aligned}
&\mathbb{E}\left[\left|(1-\Delta_{\mathfrak{g}})^{-\frac{\varepsilon}{2}}H_k\left(\mathbf{P}_N\Psi(\omega,t,\mathbf{x});\sigma_N(t,\mathbf{x})\right)\right|^2\right]\\
&\quad=\sum_{n,n'\geq 0}\frac{\varphi_n(\mathbf{x})}{\langle\lambda_n\rangle^{\varepsilon}}\frac{\varphi_{n'}(\mathbf{x})}{\langle\lambda_{n'}\rangle^{\varepsilon}}\int_{\mathcal{M}}\int_{\mathcal{M}}\mathbb{E}\big[H_k\left(\mathbf{P}_N\Psi(\omega,t,\mathbf{x}_1);\sigma_N(t,\mathbf{x}_1)\right)\\
&\qquad\qquad\times H_k\left(\mathbf{P}_N\Psi(t,\mathbf{x}_2);\sigma_N(t,\mathbf{x}_2)\right)\big]\varphi_n(\mathbf{x}_1)\varphi_{n'}(\mathbf{x}_2)d\mathbf{x}_1d\mathbf{x}_2.
\end{aligned}$$

Now we use (3.1), hence

$$\begin{aligned}\mathbb{E}\Big[\big|(1-\Delta_{\mathfrak{g}})^{-\frac{\varepsilon}{2}}H_k\big(\mathbf{P}_N\Psi(\omega,t,\mathbf{x});\sigma_N(t,\mathbf{x})\big)\big|^2\Big]&\\ = k!\sum_{n,n'\geq 0}\frac{\varphi_n(\mathbf{x})}{\langle\lambda_n\rangle^{\varepsilon}}\frac{\varphi_{n'}(\mathbf{x})}{\langle\lambda_{n'}\rangle^{\varepsilon}}\int_{\mathcal{M}^2}\big[\gamma_N^t(\mathbf{x}_1,\mathbf{x}_2)\big]^k d\mathbf{x}_1 d\mathbf{x}_2&\\ = k!\Big((1-\Delta_{\mathfrak{g}})_{\mathbf{x}_1}^{-\frac{\varepsilon}{2}}(1-\Delta_{\mathfrak{g}})_{\mathbf{x}_2}^{-\frac{\varepsilon}{2}}\big[\gamma_N^t(\mathbf{x}_1,\mathbf{x}_2)\big]^k\Big)\big|_{\mathbf{x}_1=\mathbf{x}_2=\mathbf{x}},&\end{aligned}$$

where we define

$$\begin{aligned}\gamma_N^t(\mathbf{x}_1,\mathbf{x}_2)&\stackrel{\text{def}}{=}\mathbb{E}\big[\mathbf{P}_N\Psi(\omega,t,\mathbf{x}_1)\cdot\mathbf{P}_N\Psi(\omega,t,\mathbf{x}_2)\big]\\ &=\sum_{n\geq 0}\psi_0^2\left(N^{-2}\lambda_n^2\right)\left(\int_0^t\left[\frac{\sin\big((t-t')\langle\lambda_n\rangle\big)}{\langle\lambda_n\rangle}\right]^2 dt'\right)\varphi_n(\mathbf{x}_1)\varphi_n(\mathbf{x}_2),\end{aligned}$$

the last equality resulting from Itô's isometry. In particular, in view of the second line in (1.21), we see that γ_N^t can be decomposed as

$$\gamma_N^t=\frac{t}{2}\gamma_N+\widetilde{\gamma}_N^t,$$

where γ_N is given in (3.10), and

$$\widetilde{\gamma}_N^t(\mathbf{x}_1,\mathbf{x}_2)=-\sum_{n\geq 0}\psi_0^2\left(N^{-2}\lambda_n^2\right)\frac{\sin\left(2t\langle\lambda_n\rangle\right)}{4\langle\lambda_n\rangle^3}\varphi_n(\mathbf{x}_1)\varphi_n(\mathbf{x}_2).$$

Hence, using the product estimate of Corollary 2.7 (854), we get for any $t\in[0,T]$

$$\begin{aligned}\Big\|\big(\gamma_N^t\big)^k\Big\|_{B_{\infty,\infty}^{-\frac{\varepsilon}{2}}(\mathcal{M})\times B_{\infty,\infty}^{-\frac{\varepsilon}{2}}(\mathcal{M})}&\lesssim_T\sum_{\ell=0}^k\Big\|\gamma_N^\ell\big(\widetilde{\gamma}_N^t\big)^{k-\ell}\Big\|_{B_{\infty,\infty}^{-\frac{\varepsilon}{2}}(\mathcal{M})\times B_{\infty,\infty}^{-\frac{\varepsilon}{2}}(\mathcal{M})}\\ &\lesssim\sum_{\ell=0}^k\big\|\gamma_N^\ell\big\|_{B_{\infty,\infty}^{-\frac{\varepsilon}{2}}(\mathcal{M})\times B_{\infty,\infty}^{-\frac{\varepsilon}{2}}(\mathcal{M})}\big\|\widetilde{\gamma}_N^t\big\|_{B_{\infty,\infty}^{\varepsilon}(\mathcal{M})\times B_{\infty,\infty}^{\varepsilon}(\mathcal{M})}^{k-\ell}.\end{aligned}$$

Now from Proposition 3.6, we have that $\|\gamma_N^\ell\|_{W^{-\frac{\varepsilon}{2},-\frac{\varepsilon}{2},\infty}}$ is bounded uniformly in N for any $\varepsilon>0$. As for the other term, we can estimate it directly with

Cauchy–Schwarz inequality and Lemma 2.1:

$$
\begin{aligned}
&\big\|\widetilde{\gamma}_N^t\big\|_{B^{\varepsilon}_{\infty,\infty}(\mathcal{M})\times B^{\varepsilon}_{\infty,\infty}(\mathcal{M})}\\
&= \sup_{M_1,M_2\in 2^{\mathbb{Z}_+}} M_1^{\varepsilon}M_2^{\varepsilon}\\
&\quad\times \sup_{\mathbf{x}_1,\mathbf{x}_2\in\mathcal{M}} \left|\sum_{n\geq 0}\psi_0\left(N^{-2}\lambda_n^2\right)\psi_{M_1^2}\left(\lambda_n^2\right)\psi_{M_2^2}\left(\lambda_n^2\right)\frac{\sin\left(2t\langle\lambda_n\rangle\right)}{4\langle\lambda_n\rangle^3}\varphi_n(\mathbf{x}_1)\varphi_n(\mathbf{x}_2)\right|\\
&\lesssim \sup_{M_1,M_2\in 2^{\mathbb{Z}_+}} M_1^{\varepsilon}M_2^{\varepsilon}\\
&\quad\times \sup_{\mathbf{x}_1,\mathbf{x}_2\in\mathcal{M}} \left(\sum_{n\geq 0}\psi_0\left(N^{-2}\lambda_n^2\right)^2\psi_{M_1^2}\left(\lambda_n^2\right)\psi_{M_2^2}\left(\lambda_n^2\right)\frac{\sin\left(2t\langle\lambda_n\rangle\right)^2}{\langle\lambda_n\rangle^3}\varphi_n(\mathbf{x}_1)^2\right)^{\frac{1}{2}}\\
&\quad\times \left(\sum_{n\geq 0}\psi_{M_1^2}\left(\lambda_n^2\right)\psi_{M_2^2}\left(\lambda_n^2\right)\frac{1}{\langle\lambda_n\rangle^3}\varphi_n(\mathbf{x}_2)^2\right)^{\frac{1}{2}}\\
&\lesssim \sup_{M_1\sim M_2\lesssim N} M_1^{2\varepsilon}\sum_{n\geq 0}\psi_{M_1^2}\left(\lambda_n^2\right)\psi_{M_2^2}(\lambda_n^2)\frac{1}{\langle\lambda_n\rangle^3}\\
&\lesssim \sup_{M_1\lesssim N} M_1^{2\varepsilon-1}\leq C<+\infty
\end{aligned}
$$

uniformly in $N\in\mathbb{N}$.

Thus we can conclude as in the proof of Proposition 3.4 that $\mathbb{E}|(1-\Delta_{\mathfrak{g}})^{-\varepsilon}:\Psi_N^k(t,x):|^2$ is uniformly bounded in N, from which we get a uniform bound in

$$L^p\left(\Omega;L^q([0,T];W^{-\varepsilon,\infty}(\mathcal{M}))\right)$$

for any $1\leq p,q<\infty$.

As for the convergence of $H_k(\mathbf{P}_N\Psi(\omega,t,\mathbf{x});\sigma_N(t,\mathbf{x}))$, we have again for $N_1<N_2$

$$
\begin{aligned}
\mathbb{E}\Big|(1-\Delta_{\mathfrak{g}})^{-\frac{\varepsilon}{2}}&\Big[H_k\big(\mathbf{P}_{N_1}\Psi(\omega,t,\mathbf{x});\sigma_{N_1}(t,\mathbf{x})\big)-H_k\big(\mathbf{P}_{N_2}\Psi(\omega,t,\mathbf{x});\sigma_{N_2}(t,\mathbf{x})\big)\Big]\Big|^2\\
&= k!(1-\Delta_{\mathfrak{g}})_{\mathbf{x}_1}^{-\frac{\varepsilon}{2}}(1-\Delta_{\mathfrak{g}})_{\mathbf{x}_2}^{-\frac{\varepsilon}{2}}\Big[\big(\mathbf{P}_{N_1}^2\gamma^t(\mathbf{x}_1,\mathbf{x}_2)\big)^k\\
&\qquad-2\big(\mathbf{P}_{N_1}\mathbf{P}_{N_2}\gamma^t(\mathbf{x}_1,\mathbf{x}_2)\big)^k+\big(\mathbf{P}_{N_2}^2\gamma^t(\mathbf{x}_1,\mathbf{x}_2)\big)^k\Big]_{\big|\mathbf{x}_1=\mathbf{x}_2=\mathbf{x}}\\
&\lesssim \Big\|\big(\mathbf{P}_{N_1}^2\gamma^t(\mathbf{x}_1,\mathbf{x}_2)\big)^k-2\big(\mathbf{P}_{N_1}\mathbf{P}_{N_2}\gamma^t(\mathbf{x}_1,\mathbf{x}_2)\big)^k\\
&\qquad+\big(\mathbf{P}_{N_2}^2\gamma^t(\mathbf{x}_1,\mathbf{x}_2)\big)^k\Big\|_{B^{-\frac{\varepsilon}{2}}_{\infty,\infty}(\mathcal{M})\times B^{-\frac{\varepsilon}{2}}_{\infty,\infty}(\mathcal{M})}.
\end{aligned}
$$

Writing as above $\mathbf{P}_{N_2}^2\gamma^t=\frac{t}{2}\mathbf{P}_{N_2}^2\gamma+\mathbf{P}_{N_2}^2\widetilde{\gamma}^t$, we then estimate for $t\in[0,T]$ the

contribution of

$$
\begin{aligned}
&\Big\|\big(\mathbf{P}_{N_2}^2\gamma^t(\mathbf{x}_1,\mathbf{x}_2)\big)^k-\big(\mathbf{P}_{N_1}\mathbf{P}_{N_2}\gamma^t(\mathbf{x}_1,\mathbf{x}_2)\big)^k\Big\|_{B^{-\frac{\varepsilon}{2}}_{\infty,\infty}(\mathcal{M})\times B^{-\frac{\varepsilon}{2}}_{\infty,\infty}(\mathcal{M})}\\
&\lesssim\sum_{\ell=0}^{k}\Big\|\big(\mathbf{P}_{N_2}^2\gamma(\mathbf{x}_1,\mathbf{x}_2)\big)^{\ell}\big(\mathbf{P}_{N_2}^2\widetilde{\gamma}^t(\mathbf{x}_1,\mathbf{x}_2)\big)^{k-\ell}\\
&\qquad-\big(\mathbf{P}_{N_1}\mathbf{P}_{N_2}\gamma(\mathbf{x}_1,\mathbf{x}_2)\big)^{\ell}\big(\mathbf{P}_{N_1}\mathbf{P}_{N_2}\widetilde{\gamma}^t(\mathbf{x}_1,\mathbf{x}_2)\big)^{k-\ell}\Big\|_{B^{-\frac{\varepsilon}{2}}_{\infty,\infty}(\mathcal{M})\times B^{-\frac{\varepsilon}{2}}_{\infty,\infty}(\mathcal{M})}\\
&\lesssim\sup_{\ell\leq k}\Big\|\Big[\big(\mathbf{P}_{N_2}^2\gamma(\mathbf{x}_1,\mathbf{x}_2)\big)^{\ell}-\big(\mathbf{P}_{N_1}\mathbf{P}_{N_2}\gamma(\mathbf{x}_1,\mathbf{x}_2)\big)^{\ell}\Big]\\
&\qquad\qquad\big(\mathbf{P}_{N_2}^2\widetilde{\gamma}^t(\mathbf{x}_1,\mathbf{x}_2)\big)^{k-\ell}\Big\|_{B^{-\frac{\varepsilon}{2}}_{\infty,\infty}(\mathcal{M})\times B^{-\frac{\varepsilon}{2}}_{\infty,\infty}(\mathcal{M})}\\
&+\Big\|\big(\mathbf{P}_{N_1}\mathbf{P}_{N_2}\gamma(\mathbf{x}_1,\mathbf{x}_2)\big)^{\ell}\Big[\big(\mathbf{P}_{N_2}^2\widetilde{\gamma}^t(\mathbf{x}_1,\mathbf{x}_2)\big)^{k-\ell}\\
&\qquad\qquad-\big(\mathbf{P}_{N_1}\mathbf{P}_{N_2}\widetilde{\gamma}^t(\mathbf{x}_1,\mathbf{x}_2)\big)^{k-\ell}\Big]\Big\|_{B^{-\frac{\varepsilon}{2}}_{\infty,\infty}(\mathcal{M})\times B^{-\frac{\varepsilon}{2}}_{\infty,\infty}(\mathcal{M})}\\
&\lesssim\sup_{\ell\leq k}\big(\mathrm{I}+\mathrm{II}\big).
\end{aligned}
$$

Using again the product estimate of Corollary 2.7 (854), we bound

$$
\begin{aligned}
\mathrm{I}&\lesssim\Big\|\big(\mathbf{P}_{N_2}^2\gamma(\mathbf{x}_1,\mathbf{x}_2)\big)^{\ell}-\big(\mathbf{P}_{N_1}\mathbf{P}_{N_2}\gamma(\mathbf{x}_1,\mathbf{x}_2)\big)^{\ell}\Big\|_{B^{-\frac{\varepsilon}{2}}_{\infty,\infty}(\mathcal{M})\times B^{-\frac{\varepsilon}{2}}_{\infty,\infty}(\mathcal{M})}\\
&\qquad\times\Big\|\mathbf{P}_{N_2}^2\widetilde{\gamma}^t(\mathbf{x}_1,\mathbf{x}_2)\Big\|^{k-\ell}_{B^{\varepsilon}_{\infty,\infty}(\mathcal{M})\times B^{\varepsilon}_{\infty,\infty}(\mathcal{M})}\\
&\lesssim N_1^{-\widetilde{\varepsilon}},
\end{aligned}
$$

for some $0<\widetilde{\varepsilon}\ll\varepsilon$. This follows from (3.13)-(3.14) similarly as in the proof of Proposition 3.4, along with the previous bound on $\widetilde{\gamma}^t$. As for II, we use also the product estimate to get

$$
\begin{aligned}
\mathrm{II}\lesssim&\Big\|\big(\mathbf{P}_{N_1}\gamma(\mathbf{x}_1,\mathbf{x}_2)\big)^{\ell}\Big\|_{B^{-\frac{\varepsilon}{2}}_{\infty,\infty}(\mathcal{M})\times B^{-\frac{\varepsilon}{2}}_{\infty,\infty}(\mathcal{M})}\\
&\times\Big\|\big(\mathbf{P}_{N_2}^2\widetilde{\gamma}^t(\mathbf{x}_1,\mathbf{x}_2)\big)^{k-\ell}-\big(\mathbf{P}_{N_1}\mathbf{P}_{N_2}\widetilde{\gamma}^t(\mathbf{x}_1,\mathbf{x}_2)\big)^{k-\ell}\Big\|_{B^{\varepsilon}_{\infty,\infty}(\mathcal{M})\times B^{\varepsilon}_{\infty,\infty}(\mathcal{M})},
\end{aligned}
$$

and we can gain a small negative power of N_1 in the second term by proceeding as for the bound on $\widetilde{\gamma}^t$ above and using that the supremum of $M_1^{2\varepsilon-1}$ now runs over $N_1\lesssim M_1\lesssim N_2$. The second contribution $(\mathbf{P}_{N_1}^2\gamma^t(\mathbf{x}_1,\mathbf{x}_2))^k-(\mathbf{P}_{N_1}\mathbf{P}_{N_2}\gamma^t(\mathbf{x}_1,\mathbf{x}_2))^k$ is estimated similarly. This shows that $\{H_k(\mathbf{P}_N\Psi(\omega,t,\mathbf{x});\sigma_N(t,\mathbf{x}))\}_{N\in\mathbb{N}}$ is a Cauchy sequence in $L^p(\Omega;L^q([0,T];W^{-\varepsilon,\infty}(\mathcal{M})))$ for any finite $p,q\geq 1$.

Let us finally turn to the continuity property of Ψ_N and Ψ. As in the

previous section, we compute for any $h, t \in [0,T]$ and $\mathbf{x} \in \mathcal{M}$

$$\begin{aligned}
&\mathbb{E}\big|(1-\Delta_{\mathfrak{g}})^{-\varepsilon}(\tau_h\Psi-\Psi)(\omega,t,\mathbf{x})\big|^2 \\
&\qquad = \sum_{n\geq 0}\frac{\varphi_n(\mathbf{x})^2}{\langle\lambda_n\rangle^{2\varepsilon}}\left\{\int_t^{t+h}\left[\frac{\sin\big((t+h-t')\langle\lambda_n\rangle\big)}{\langle\lambda_n\rangle}\right]^2 dt'\right. \\
&\qquad\quad \left. + \int_0^t\left[\frac{\sin\big((t+h-t')\langle\lambda_n\rangle\big)-\sin\big((t-t')\langle\lambda_n\rangle\big)}{\langle\lambda_n\rangle}\right]^2 dt'\right\} \\
&\qquad \lesssim \sum_{n\geq 0}\frac{\varphi_n(\mathbf{x})^2}{\langle\lambda_n\rangle^{2+2\varepsilon}}\left\{h + t\sin\left(\frac{h\langle\lambda_n\rangle}{2}\right)^2\right\} \\
&\qquad \lesssim \sum_{n\geq 0}\frac{\varphi_n(\mathbf{x})^2}{\langle\lambda_n\rangle^{2+2\varepsilon}}(h\langle\lambda_n\rangle)^{\varepsilon} \lesssim h^{\varepsilon},
\end{aligned}$$

which leads as in the previous section to $\Psi \in C([0,T];W^{-\varepsilon,\infty}(\mathcal{M}))$ almost surely.

Lastly, the tail estimate is obtained through the same argument as in the previous section. This concludes the proof of Proposition 0.1. □